An Introduction to Ergodicity Economics

Ole Peters and Alexander Adamou

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Long-term growth

$\boxed{\text{Axiom: Agents maximize wealth.}}$

<u>Twist:</u> not expected wealth but real wealth over time.

Interesting because

Ergodicity is broken
$\rightarrow$ expected wealth $\neq$ real wealth over time.

Why the fuss?

"The economy is not ergodic."

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Look at *standard models* in economics, find ergodicity broken *there*.

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Ergodicity = property of mathematical models.

Look at *standard models* in economics, find ergodicity broken *there*.

Implication: standard analyses need re-doing.
$\rightarrow$ Different worldview.

Coin tosses

- The infamous coin toss
- The leveraged coin toss
- The cooperating coin toss
- The social coin toss

Coin tosses

- **The infamous coin toss**
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The infamous coin toss

Let us define our model of the dynamics of your wealth under the rules we just specified.[1] At regular intervals of duration δt, we randomly generate a factor r with each possible value $r_i \in \{0.6, 1.5\}$ occurring with probability 1/2,

$$r = \begin{cases} 0.6 & \text{with probability } 1/2 \\ 1.5 & \text{with probability } 1/2 \end{cases} \tag{1.1}$$

and multiply current wealth by that factor, so that

$$x(t + \delta t) = x(t)r. \tag{1.2}$$

The infamous coin toss

Two properties:

The infamous coin toss

Two properties:

- 1. Expected value
  $\langle x(t+1)\rangle = \frac{1}{2}\left(r_H x(t) + r_T x(t)\right) = 1.05x(t)$.
  By induction, $\langle x(t)\rangle = x(0) \times 1.05^t$.
  Per-round growth $\underline{g_{\langle\rangle} = 1.05.}$

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- 2. Time-average growth
  $x(t) = x(0)\prod_{\tau=1}^{t} r_\tau = x(0) r_H^{n_H} r_T^{n_T}$.
  per-round growth $g = \left(\frac{x(t)}{x(0)}\right)^{1/t} = r_H^{n_H/t} r_T^{n_T/t}$
  long-time limit $\lim_{t\to\infty} r_H^{n_H/t} r_T^{n_T/t} = r_H^{1/2} r_T^{1/2}$
  $\underline{g_t \approx 0.949}$

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Comments:

- discrete form of Geometric Brownian Motion (GBM): applies to all of finance theory.

The infamous coin toss

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- discrete form of Geometric Brownian Motion (GBM): applies to all of finance theory.
- multiplicativity: evolution, life, investing, learning, growing.

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Averages

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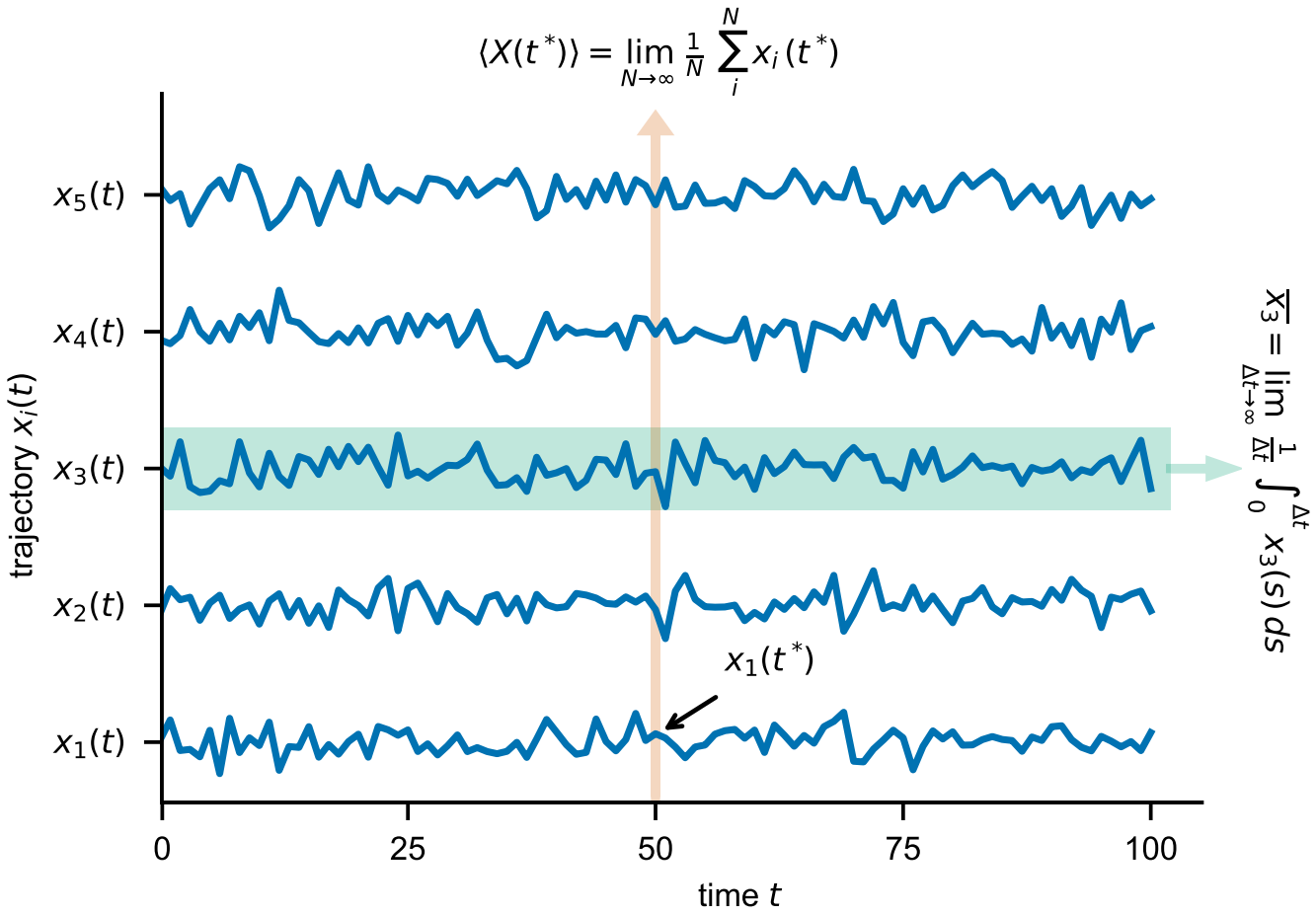

Chapter 2, p.9:

Ergodicity

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Philosophy: keep the text light.

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Simple ergodicity problem:

$$\text{expected value} \neq \text{time average}.$$

Ergodicity

Philosophy: keep the text light.

Simple ergodicity problem:

$$\text{expected value} \neq \text{time average}.$$

But $\approx$150 years of theory behind this.

Dynamical systems

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Dynamical sytem: (X, F, μ, T)

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$X =$ all possible identities x_i (you, me; this dice, that dice...).

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$X =$ all possible identities x_i (you, me; this dice, that dice...).

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$T =$ time passing. Identities change according to $T^n(A)$. These changes don't affect the size of the collection, $\mu(T^{-1}A) = \mu(A)$.

Dynamical systems

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- Mathematically, each entity marches through time deterministically.
- But that's not physically visible.
- Think of x_i as deeply hidden and unobservable, "the true you."
- We don't know the true you, but we can measure some things "how tall are you?"

Ergodic hierarchy

Ergodicity means

- Metric transitivity

Ergodic hierarchy

Ergodicity means

- Metric transitivity
  "any differences are transitory."
  For any sets $A, B \in F$, there exists an n such that $\mu(T^{-n}(A) \cap B) > 0$ (mixing).

  Equivalently, "everything always changes.'
  If A satisfies $T^{-1}(A) = A$, then either $\mu(A) = 1$ or $\mu(A) = 0$ (only trivial invariant sets).

Ergodic hierarchy

Ergodicity means

- Metric transitivity

$\rightarrow$ Equality of distributions

Ergodic hierarchy

Ergodicity means

- Metric transitivity

$\rightarrow$ Equality of distributions
"In the long run, measurements in any two systems show the same values equally often."

Ergodic hierarchy

Ergodicity means

- Metric transitivity

$\rightarrow$ Equality of distributions

$\rightarrow$ Equality of averages

Ergodic hierarchy

Ergodicity means

- Metric transitivity

$\rightarrow$ Equality of distributions

$\rightarrow$ Equality of averages "Averaging over time is equivalent to averaging over the ergodic ensemble."

Ergodic hierarchy

Ergodicity means

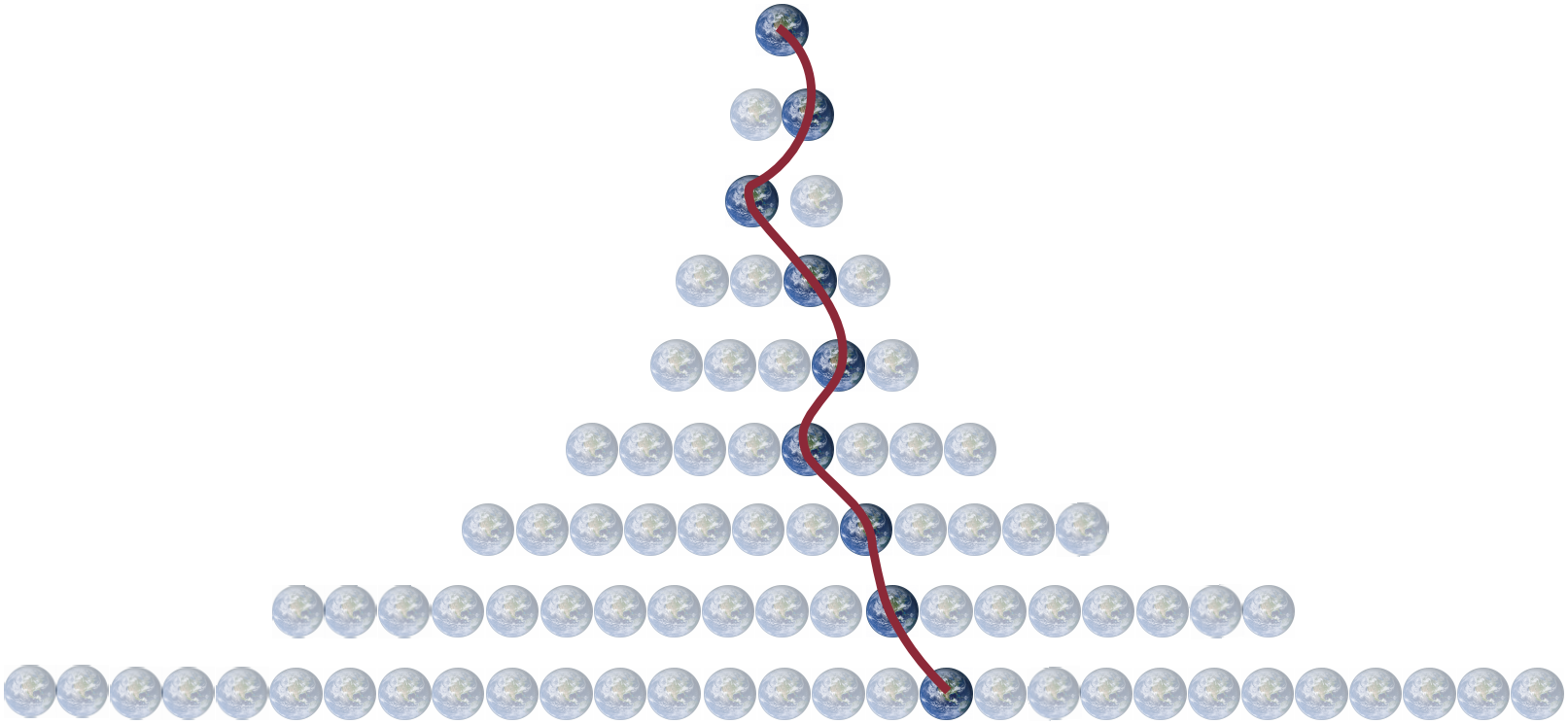
- Metric transitivity

$\rightarrow$ Equality of distributions

$\rightarrow$ Equality of averages

Chapter 4, p.34: Models

More tools.

- Brownian motion
- Stochastic differential equations
- Itô calculus
- Geometric Brownian motion

Thank you.